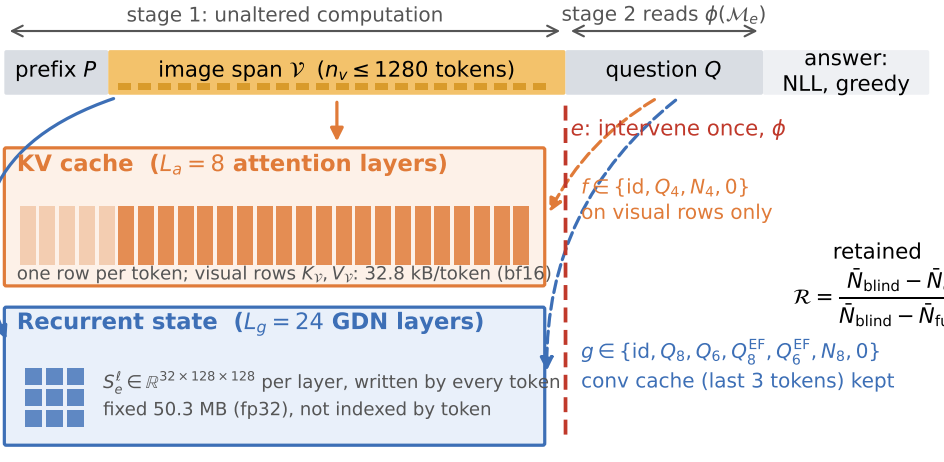
stage 1: unaltered computation
stage 2 reads $\phi(\mathcal{M}_e)$
prefix P
image span $\mathcal{V}$ ($n_v \le 1280$ tokens)
question Q
answer: NLL, greedy
e: intervene once, ϕ
$f \in \{\mathrm{id}, Q_4, N_4, 0\}$
on visual rows only
KV cache ($L_a = 8$ attention layers)
one row per token; visual rows K_v, V_v: 32.8 kB/token (bf16)
retained
$\mathcal{R} = \frac{\bar{N}_{\mathrm{blind}} - \bar{N}_c}{\bar{N}_{\mathrm{blind}} - \bar{N}_{\mathrm{full}}}$
Recurrent state ($L_g = 24$ GDN layers)
$S_e^\ell \in \mathbb{R}^{32\times128\times128}$ per layer, written by every token
fixed 50.3 MB (fp32), not indexed by token
$g \in \{\mathrm{id}, Q_8, Q_6, Q_8^{\mathrm{EF}}, Q_6^{\mathrm{EF}}, N_8, 0\}$
conv cache (last 3 tokens) kept